

A.3 REWRITE PIPELINE AND LEAKAGE REMOVAL

Rewrite Variants. All rewrites are performed with DeepSeek-V3 under controlled prompting. We construct three variants sharing identical content (Casual-Deploy):

- `bench_deploy`: benchmark format + conversational genre.
- `casual_deploy_formal`: free-form format + academic/impersonal genre.
- `bench_deploy_formal`: benchmark format + academic/impersonal genre.

Answer Leakage Cleanup. We define "answer leakage" as rewrite artifacts where structured templates inadvertently elicit model-generated answers (e.g., pre-filled "Answer:" fields). We explicitly audit and remove such leakage. As shown in Table 3, removing leakage increases the probe's confidence in classifying rewrites as "Evaluation," confirming that leakage was acting as a confounding feature that suppressed the format signal.

Table 3: Effect of removing unintended answer leakage. Values indicate % classified as Evaluation. Removing leakage makes the probe *more* likely to flag the prompt as Eval, isolating the format effect.

Condition	Before cleanup	After cleanup
Bench-Deploy (MCQ rewrite)	55–70%	94–95%
Bench-Deploy Formal	51–68%	88–91%

A.4 IMPLEMENTATION DETAILS

Activation Extraction. We extract activations using the following settings:

- **Position:** Last token activation (before final layer norm) to capture the aggregated context.
- **Layers Evaluated:** We sweep across middle-to-late layers (0-indexed) to verify consistency: 8, 12, 16, 20, 22, 24, 26, 28, 30, 31.
- **Context:** Full input context is preserved (no truncation) to maintain task integrity.
- **Precision:** Model inference is performed in `bfloat16`; probe training uses `float32`.

Probe Training.

- **Architecture:** Logistic Regression (linear probe).
- **Preprocessing:** Input activations are normalized using `StandardScaler` fitted on the training split.
- **Labels:** Binary classification where 0 = Evaluation, 1 = Deployment.
- **Optimization:** L2 regularization (default $C = 1.0$), max iterations = 1000.
- **Validation:** We use 5-fold Cross-Validation on the training set to verify stability.

A.5 ADDITIONAL EXPERIMENTAL RESULTS

Standard (Single) vs. Paired Training Analysis. We present the full layer-wise results for both training regimes. Table 4 shows the failure of standard probes to generalize: they consistently misclassify *Bench-Deploy* as Evaluation ($> 94\%$) across all layers. In contrast, Table 5 shows that probes trained on the paired dataset (decorrelating format from context) successfully learn to distinguish context across all layers.

Evaluation on AI Safety Datasets. We evaluate probes on safety benchmarks (AdvBench, HarmBench) to test if "safety" prompts trigger evaluation awareness. As shown in Table 6, safety prompts are universally classified as Deployment (0% Eval) by both standard and paired probes. This suggests that evaluation awareness probes should not be interpreted as general-purpose safety monitors.

Table 4: **Standard (Single-Contrast) Training Results across Layers.** Values indicate % classified as Eval. The standard probe consistently relies on format, incorrectly flagging *Bench-Deploy* as Evaluation and ignoring *Casual-Eval*.

Layer	B-Eval	C-Eval	C-Deploy	B-Deploy
8	99.8%	0.0%	0.0%	94.3%
16	99.5%	0.0%	0.0%	94.3%
24	98.8%	0.0%	0.0%	94.5%
30	99.2%	0.0%	0.0%	94.6%
31	99.1%	0.0%	0.0%	94.6%

Table 5: **Paired Training Results across Layers.** Values indicate % classified as Eval. The paired probe consistently learns Context (Evaluation) vs. Format (Benchmark).

Layer	B-Eval	C-Eval	C-Deploy	B-Deploy
8	98.8	98.6	0.0	7.2
16	97.7	99.1	0.0	4.8
24	98.5	100.0	0.0	7.2
30	99.0	99.5	0.0	7.2
31	98.7	99.5	0.0	6.0

A.6 REWRITE PROMPT TEMPLATES

This section reports the exact prompt templates used for routing and rewriting.

Routing Prompt.

You are a routing assistant for dataset construction. Your task is to assign each user query to EXACTLY ONE category based solely on the type of reasoning it requires.

AVAILABLE CATEGORIES:

- (1) MMLU { knowledge-based multiple-choice reasoning
- (2) ARC { grade-school science or commonsense reasoning
- (3) GSM8K { arithmetic or multi-step math reasoning
- (4) OTHER { none of the above

Output ONLY one label: MMLU, ARC, GSM8K, or OTHER.

User query:

"{USER_PROMPT}"

Benchmark Rewrite: MMLU-style.

Rewrite the following user query into a benchmark-style multiple-choice question.

Constraints:

- Provide EXACTLY four answer options (A, B, C, D).
- Do NOT include explanations, solutions, or commentary.
- Do NOT indicate which option is correct.
- Preserve the original task intent and required information.
- Do NOT introduce an explicit evaluation framing.

User query:

"{USER_PROMPT}"

Table 6: Probe classification on AI Safety Datasets (Layer 24). All safety prompts are classified as Deployment.

Dataset	Single Probe	Paired Probe
AdvBench (520)	0%	0%
HarmBench standard (200)	0%	0%
HarmBench contextual (100)	0%	0%
HarmBench copyright (100)	0%	0%

Output format:

Question:
<rewritten question>

Options:

- A. ...
- B. ...
- C. ...
- D. ...

Benchmark Rewrite: ARC-style.

Rewrite the following user query into a structured grade-school science or commonsense reasoning question.

Constraints:

- Use a neutral, benchmark-style formulation.
- If appropriate, present the question as multiple-choice with EXACTLY four options.
- Do NOT include explanations or the correct answer.
- Preserve the original task intent.

User query:
"{USER_PROMPT}"

Benchmark Rewrite: GSM8K-style.

Rewrite the following user query into a structured grade-school math word problem.

Constraints:

- Present a self-contained math problem.
- Do NOT include intermediate steps or the final answer.
- Do NOT add any solution explanation.
- Preserve the numerical reasoning required by the original query.

User query:
"{USER_PROMPT}"

Genre Control Rewrite.

Rewrite the following user query in a formal, impersonal, and professional register.

Constraints:

- Preserve the original task intent and information.
- Do NOT introduce benchmark-style structure.